

Nitrogen Reservoirs in Fulgurites

A Provenance Protocol and the Case Against Structural Si–N in Lightning Glass

Contract

This is the chemistry lane. It does not claim a coupled carbon–nitrogen–silicon system, a new mineral, a planetary circuit, or a mystical archive. It treats fulgurites as geochemical materials formed by lightning and asks a narrow, testable question:

«When nitrogen is detected in a fulgurite, what reservoir is it actually in?»

The answer matters because the sentence “there is nitrogen in the glass” can mean at least four different things: nitrogen bonded into the silica network, nitrogen trapped as molecular gas inside sealed vesicles, nitrogen adsorbed onto surfaces or weathering rinds, or nitrogen contained in organic residues. These are not the same claim. They do not require the same evidence. They do not carry the same meaning.

The operating rule is simple:

«Resolve the reservoir before naming the nitrogen.

No reservoir, no claim.»

This follows the project’s No-Crown rule: mythic labels may generate questions, but they do not support conclusions; the evidence strand cannot borrow authority from the narrative strand.

1. Abstract

Fulgurites are natural glasses formed when cloud-to-ground lightning fuses sand, soil, rock, or related substrates. Pasek, Block, and Pasek describe fulgurites as natural glasses produced by lightning and classify their morphologies across sand, clay, caliche, rock, and related forms.

This write-up argues that structural nitrogen, meaning nitrogen bonded directly into the silica network as Si–N or Si–O–N, should not be expected in ordinary terrestrial fulgurite glass. The reason is mechanistic: ordinary fulgurites form in air, in oxidizing conditions, and quench rapidly. Experimental work on nitrogen in silicate melts shows that nitrogen behavior is strongly controlled by oxygen fugacity; over broad conditions N_2 is the dominant dissolved species, while N^{3-} -type structural species become significant only under reducing conditions.

The useful chemistry contribution is therefore not “we found structural nitrogen.” The stronger contribution is a provenance protocol: separate nitrogen into lattice, gas, surface, and organic reservoirs; test each with the right instrument; state an upper bound when the bonding test is

mute; and avoid false positives from surface contamination, organic matter, SIMS recombination, or modern air infiltration.

The two most promising positive archive lanes are: first, sealed-vesicle gas, because Navarro-González and colleagues reported CO₂, CO, and NO in glassy bubbles from a Late Pleistocene Libyan Desert fulgurite; and second, carbon-correlated reduced phases, because lightning has been shown to reduce phosphate in fulgurites and iron silicides have been reported and reviewed in fulgurite contexts.

The expected output is therefore:

1. an upper bound or negative result for structural Si-N;
2. provenance-resolved accounting of any nitrogen detected;
3. replication/extension of sealed-vesicle gas archive signals;
4. spatial testing of carbon-correlated reduced phosphorus and iron-silicide phases;
5. a strictly controlled pilot on post-strike bioreceptivity, kept separate from clean nitrogen chemistry.

2. Background: What Fulgurites Are

A fulgurite is a natural melt product produced when lightning delivers enough energy into a substrate to fuse, vitrify, and sometimes vaporize mineral grains. In silica-rich settings, the resulting material is commonly dominated by lechatelierite, amorphous silica glass. The morphology and chemistry of a fulgurite depend on the struck substrate, the discharge conditions, moisture, organic content, metal contamination, and cooling history.

This matters because fulgurites are not uniform laboratory glasses. They are small, violent, heterogeneous melt products. A single specimen can contain clean silica glass, mixed groundmass, bubbles, charred residues, metallic inclusions, weathered exterior surfaces, and cracks that later admit fluids or contaminants. That heterogeneity is exactly why nitrogen provenance must be resolved before interpretation.

The chemistry question is not simply:

«Is there nitrogen?»

The better question is:

«Where is the nitrogen, what chemical form is it in, and how do we know it belongs to the lightning event rather than to later contamination or biological/weathering overprint?»

3. The Actual Gap

Fulgurites can preserve event information. That is not the speculative part.

The sealed-vesicle gas result is already real enough to treat as a target for replication: Navarro-González and colleagues reported CO₂, CO, and NO trapped inside glassy bubbles in a Libyan Desert fulgurite and used those gases, plus isotopic and C/N information, to infer paleoecological context.

Lightning-driven redox chemistry is also a real lane. Pasek and Block reported that cloud-to-ground lightning can reduce phosphate in fulgurites, with reduced phosphorus found in several specimens. Iron silicides have also been reviewed in fulgurite occurrences, with the key issue being the extremely reducing microenvironments required for their formation.

The gap is more specific:

«Positive nitrogen signals in fulgurites are easy to overinterpret unless they are assigned to the correct reservoir.»

A nitrogen peak, by itself, is not a structural claim. A surface nitrate signal is not event nitrogen. A CN signal is not automatically lattice nitrogen. A SiN⁻ secondary ion in SIMS is not automatically proof of a pre-existing Si–N bond. A bubble gas signal is not the same as a bonded network signal.

The chemistry lane is therefore a sorting machine.

4. Why Structural Si–N Is Not Expected

Structural Si–N in ordinary fulgurite glass is a predicted negative.

That is the central chemistry claim.

The argument has four parts.

4.1 Nitrogen speciation in silicate melts is redox-controlled

Nitrogen incorporation in silicate melts depends strongly on oxygen fugacity. Under relatively oxidizing conditions, nitrogen is generally present as molecular N₂ rather than as network-bonded nitride. Under highly reducing conditions, nitride-like N³⁻ species can become more significant and enter melt structures.

That means ordinary air-formed lightning glass is not the natural setting where one should expect structural nitride or silicon oxynitride formation.

4.2 Ordinary fulgurites form in air

Most terrestrial fulgurites form when lightning strikes sand, soil, rock, or sediment in the atmosphere. That setting is not a deliberately reducing furnace, not an ammonia treatment, and not a nitride-starting-material synthesis environment.

Even if tiny reduced microzones form locally, the broad prior remains: structural Si–N in clean silica glass should be absent or below meaningful detection.

4.3 Silicon oxynitride requires special conditions

Natural silicon oxynitride, known as sinoite, was identified as a meteoritic mineral in an enstatite chondrite, a highly reduced extraterrestrial context, not an ordinary terrestrial lightning-glass product. Confirmed terrestrial nitride minerals are exceptionally rare; siderazot, an iron nitride from a fumarolic setting, was historically reported once and only much later confirmed and structurally characterized.

The base-rate argument is important: nature can make nitrides, but ordinary struck soil is not the expected place to make structural silicon nitride or silicon oxynitride glass.

4.4 The honest expected result is an upper bound

If EELS does not see nitrogen in fresh interior glass, that does not prove “nothing interesting happened.” It means the structural-bonding claim is unsupported and should be reported as an upper bound. The chemistry document already states this as the honest default: structural Si–N is expected absent, and the value of the result is strongest when paired with provenance-resolved accounting of nitrogen in other reservoirs.

So the clean claim is:

«Structural Si–N in ordinary fulgurite glass is not expected. If nitrogen exists, the first job is to determine whether it is gas, surface, organic, or truly lattice-bound.»

5. The Four Nitrogen Reservoirs

Any nitrogen signal must be sorted into one of four reservoirs.

R1 — Lattice / Structural Nitrogen

Definition: nitrogen bonded into the silica glass network as Si–N or Si–O–N.

Prior expectation: absent or below detection in ordinary fulgurite glass.

What would confirm it: spatially coherent nitrogen in fresh interior lechatelierite, above detection, with N-K edge ELNES consistent with Si–N bonding, distinguished from nitrate, organic nitrogen, trapped gas, beam contamination, groundmass mixing, and SIMS recombination.

Status: predicted negative.

R2 — Occluded Molecular Gas

Definition: nitrogen-bearing gas trapped inside sealed vesicles, potentially including NO, NO_x, N₂ mixtures, CO, and CO₂.

Prior expectation: plausible and important.

What would confirm it: sealed-vesicle crush or stepped-heating release that differs from modern air, open-vesicle gas, rind contamination, and procedural blanks. Gas ratios, $\delta^{13}\text{C}$, $\delta^{15}\text{N}$, and where possible ^{14}C context would support provenance, but isotopes alone should not be overclaimed.

Status: plausible positive archive lane.

R3 — Surface / Weathering Nitrogen

Definition: nitrate, ammonium, adsorbed nitrogen, handling contamination, storage contamination, soil/weathering input, or nitrogen on cracks and rinds.

Prior expectation: common.

What would confirm it: exterior-to-interior gradients, concentration on cracks/rinds, disappearance in fresh interior glass, association with open surfaces, and sensitivity to cleaning/fracturing protocols.

Status: default contamination lane unless proven otherwise.

R4 — Organic Residue Nitrogen

Definition: nitrogen in charred biomass, trapped carbonaceous particles, PAHs, microbial films, bioalteration products, or soil organic matter.

Prior expectation: mixed; possible, but must be separated from lattice and gas claims.

What would confirm it: co-location with carbon phases, CN-rich signals, organic ELNES/Raman signatures, textural association with residues rather than clean lechatelierite, and separation from biological overprint.

Status: plausible but dangerous if confused with R1.

Physical entrapment is not bonding. Organic nitrogen is not lattice nitrogen. Surface nitrogen is not event nitrogen without controls. That is the whole blade.

6. Analytical Decision Tree

6.1 Sample triage

Start by separating physical domains before analysis:

1. fresh interior lechatelierite;
2. mixed groundmass;
3. sealed vesicles;
4. open vesicles;
5. weathered rind;
6. cracks and pores;
7. carbon-rich residues;
8. biologically colonized or altered zones.

Do not analyze all of these as if they are one material. A bulk result is almost guaranteed to smear reservoirs together.

6.2 NanoSIMS

NanoSIMS is useful for sensitive nitrogen detection and spatial mapping. It can map CN^- , SiN^- , NO^- , and related secondary ions across interior glass, residues, cracks, and rinds.

But NanoSIMS cannot, by itself, prove a pre-existing Si–N bond. Secondary ions can form during sputtering, and CN^- can reflect organic nitrogen, adsorbed nitrogen, or analytical recombination. The correct role of NanoSIMS is localization and reservoir targeting, not final bonding assignment.

6.3 EELS N-K ELNES

EELS is the decisive bonding-environment lane if nitrogen is present above detection. N-K edge fine structure can, in principle, distinguish bonding environments such as nitride-like, nitrate-like, and organic nitrogen.

But EELS has an honest limit: if structural nitrogen is absent or below detection, it cannot fingerprint what is not there. In the predicted default case, the result is not a dramatic positive; it is an upper bound on R1.

6.4 Vacuum crush, stepped heating, GC-MS, isotopes

This is the strongest lane for R2, the sealed-vesicle gas archive.

The ideal workflow is paired sealed-vs-open vesicle extraction, procedural blanks, modern-air references, stepped-heating release, GC-MS gas composition, and $\delta^{13}\text{C}/\delta^{15}\text{N}$ measurement. If ancient preservation is claimed, ^{14}C context becomes important.

A gas archive claim strengthens only if sealed vesicles release gas that is compositionally or isotopically distinct from modern air, soil infiltration, open vesicles, and weathered surfaces.

6.5 XANES / EXAFS for phosphorus and iron

XANES and EXAFS are appropriate for the redox archive lane, especially for phosphorus and iron oxidation states. This lane should test whether reduced P or Fe phases correlate spatially with carbon-rich zones, rather than merely identifying isolated reduced phases.

This matters because carbon can locally drive reduction in lightning melts. But that does not automatically imply nitrogen is reduced into the silica lattice. The chemistry document explicitly keeps those mechanisms separate.

6.6 Raman, XRD, SEM-EDS, microprobe

These are context tools:

- Raman: carbon phases, glass structure, possible organic/residue signatures;
- XRD: crystalline reduced phases;
- SEM-EDS: texture, inclusions, phase context;
- electron microprobe: transects across carbon, phosphorus, iron, silicon, and substrate heterogeneity.

They are powerful for mapping and selecting targets, but they do not replace direct nitrogen bonding tests.

6.7 XPS, FTIR, and surface-only methods

XPS is surface-biased and therefore dangerous unless used on freshly fractured interiors with strict controls and depth profiling. FTIR and Raman assignments for Si–N in complex glass are not decisive because vibrational features can overlap or be ambiguous.

Any surface-only nitrogen detection should be treated as R3 until proven otherwise.

6.8 Identifiability, Selectivity, and Measurement-Induced Transformation

The analytical objective is not to maximize instrument count. It is to discriminate among competing reservoir assignments with the smallest set of measurements and preparation contrasts that actually change what can be concluded. A method that reproduces an ambiguity may still be useful for quality control, but it does not add provenance information merely because it is technologically different.

6.8.1 Adaptive targeting: NanoSIMS → EELS

NanoSIMS and EELS should not be treated as two parallel votes on R1. Their strongest relationship is sequential. NanoSIMS first surveys fresh interior domains, maps nitrogen at high spatial sensitivity, and identifies candidate hotspots while excluding obvious cracks, rinds, carbon-rich residues, and mixed groundmass. EELS N-K ELNES is then targeted to the most credible high-N interior microdomains.

The operational sequence for R1 is therefore:

1. select fresh interior lechatelierite;
2. survey with NanoSIMS;
3. reject targets spatially associated with obvious R3 or R4 domains;
4. identify the highest-confidence interior-N targets;
5. interrogate those exact microdomains by EELS N-K ELNES;
6. audit preparation, beam, mounting, and background contributions;
7. report either bonding evidence that survives those controls or an upper bound on R1.

This makes NanoSIMS a targeting instrument for the conditional bonding test rather than independent proof of structural nitrogen.

6.8.2 Background nitrogen is a measured competitor, not an afterthought

In addition to R1–R4, define **B — procedural/background nitrogen** as the contribution introduced or exposed by preparation, mounting, polishing, storage, handling, extraction, analytical blanks, or laboratory environment. B is not a fifth specimen reservoir; it is a competing contribution to the observed signal and should be characterized quantitatively rather than treated only as something to subtract after the fact.

Repeated full-procedure blanks, nitrogen-free fused-silica controls, preparation replicates, and blank distributions should therefore travel with the specimen measurements. Where B is variable or comparable to the candidate event signal, the correct conclusion is limited by that uncertainty.

6.8.3 Observation is not always passive

Several methods in this program transform the material before or during observation. Crushing exposes and mixes previously sealed domains. Heating releases species and can alter or generate products. SIMS sputtering creates secondary ions and can recombine atoms in the sputter plume. These are not merely different windows onto a fixed specimen.

For every destructive or transformative method, distinguish:

- species or bonding environments demonstrably present before analysis;
- species released from a pre-existing reservoir by the analytical operation;
- species redistributed, recombined, or generated during preparation or measurement.

Method-specific controls must therefore ask not only “what signal was measured?” but also “could the operation itself have produced, transformed, or reassigned the signal?” This generalizes the existing SIMS-recombination warning to the entire analytical workflow.

6.8.4 Prefer discriminating contrasts

The design question for every added measurement is:

> **Which currently plausible reservoir assignments does this measurement separate that the existing workflow cannot?**

Provenance question	High-value discriminating contrast
R1 vs R4	clean interior vs carbon-associated domains, followed by targeted ELNES
R2 vs modern infiltration	sealed vs open vesicles plus gas ratios and blanks
R3 vs interior N	rind/crack vs fresh-interior gradients
specimen N vs B	repeated full-procedure blanks plus nitrogen-free fused silica
pre-existing vs measurement-produced species	method-specific transformation controls
R1 detectable anywhere?	NanoSIMS hotspot search → targeted EELS

Preparation contrasts may provide more provenance information than adding another instrument. Interior/exterior splits, sealed/open vesicles, clean/residue-rich domains, and controlled preparation variants directly challenge competing explanations.

6.8.5 Quantitative identifiability as a later analysis layer

Once empirical sensitivities, covariance structures, detection limits, transformation behavior, and blank distributions are available, the measurement suite can be treated quantitatively as an identifiability and experimental-design problem. Sensitivity matrices, estimability analysis, singular-value/condition analysis, profile likelihood, and optimal-design methods could then be

used to identify redundant assays, poorly constrained reservoir contrasts, and minimum discriminating measurement sets.

Those calculations are not claimed here because the required empirical inputs do not yet exist. The present protocol supplies the experimental structure needed to obtain them.

7. Sub-Study A: Sealed-Vesicle Gas Archive

The strongest positive nitrogen lane is not structural Si–N. It is sealed-vesicle gas.

Known starting point: Navarro-González and colleagues reported CO₂, CO, and NO in glass bubbles from a Late Pleistocene Libyan Desert fulgurite, using those gases to reconstruct paleoecological context.

Research question

Can sealed vesicles in multiple fulgurites preserve event-linked gas signatures distinguishable from modern air, soil infiltration, and surface contamination?

Novelty

The novelty is not “gas in one fulgurite exists.” The novelty would be:

1. multi-specimen reproducibility;
2. paired sealed-vs-open vesicle comparison;
3. stepped-heating release structure;
4. explicit modern-air and procedural blanks;
5. provenance-resolved nitrogen accounting.

Methods

- micro-CT or microscopy to identify sealed vesicles;
- vacuum crush of sealed vesicle domains;
- stepped heating;
- GC-MS for CO₂, CO, NO, N₂, O₂, Ar, and related species;
- $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ where sample size permits;
- ^{14}C context where ancient preservation is claimed;
- comparison against open vesicles and weathered rinds.

Falsifiers

The gas archive claim weakens or fails if:

- sealed-vesicle